

Synthetic Control Methods

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**Abadie, Diamond, Hainmueller
(2010)**

The Question

In 1988 California passes anti-smoking Prop 99

- increased excise tax by 25 cents per pack,
- led to indoor smoking bans in restaurants and bars city by city

What was the effect on per capita cigarette sales?

- Before and after comparisons probably confuse the impact with other changes over time...
- What is a good control for California?

Use state-level data from 1970-2000.

The Idea

Use a combination of other states to construct a **synthetic counterfactual California**.

- We observe Y_{it}, X_{it}, T_{it} .
- Assume only $i = 1$ and $t > T_0$ are **treated**.
- Construct a **donor pool** of potential controls subscripted by j .
- Choose some **weights** w_j for each entity (state) in donor pool. How?
 - Same $\mathbf{X}_1 = \sum_j w_j \mathbf{X}_j$ as treated observations (like matching).
 - Same $(Y_{1,1}, \dots, Y_{1,T_0}) = \sum_j w_j \cdot (Y_{j,1}, \dots, Y_{j,T_0})$ (like parallel trends).
 - Weights sum to one $\sum_j w_j = 1$ and maybe are non-negative (or not!)
- Idea is to match all of the X 's and all of the Y_{it} 's in the **pre-period**

Considerations

1. Potential control regions are identified:
 - Eliminate regions that have had a similar intervention
 - Remove regions where data is not available required
 - Eliminate regions with very dissimilar characteristics (avoid Biases caused by interpolation across regions with very different characteristics)
2. Assume that the first region is **uninterruptedly** exposed to the intervention of interest after some initial intervention period.
3. Balanced panel

Table 1. Cigarette sales predictor means

Variables	California		Average of 38 control states
	Real	Synthetic	
Ln(GDP per capita)	10.08	9.86	9.86
Percent aged 15–24	17.40	17.40	17.29
Retail price	89.42	89.41	87.27
Beer consumption per capita	24.28	24.20	23.75
Cigarette sales per capita 1988	90.10	91.62	114.20
Cigarette sales per capita 1980	120.20	120.43	136.58
Cigarette sales per capita 1975	127.10	126.99	132.81

NOTE: All variables except lagged cigarette sales are averaged for the 1980–1988 period (beer consumption is averaged 1984–1988). GDP per capita is measured in 1997 dollars, retail prices are measured in cents, beer consumption is measured in gallons, and cigarette sales are measured in packs.

Donor Weights

Table 2. State weights in the synthetic California

State	Weight	State	Weight
Alabama	0	Montana	0.199
Alaska	–	Nebraska	0
Arizona	–	Nevada	0.234
Arkansas	0	New Hampshire	0
Colorado	0.164	New Jersey	–
Connecticut	0.069	New Mexico	0
Delaware	0	New York	–
District of Columbia	–	North Carolina	0
Florida	–	North Dakota	0
Georgia	0	Ohio	0
Hawaii	–	Oklahoma	0
Idaho	0	Oregon	–
Illinois	0	Pennsylvania	0
Indiana	0	Rhode Island	0
Iowa	0	South Carolina	0
Kansas	0	South Dakota	0
Kentucky	0	Tennessee	0
Louisiana	0	Texas	0
Maine	0	Utah	0.334
Maryland	–	Vermont	0
Massachusetts	–	Virginia	0
Michigan	–	Washington	–
Minnesota	0	West Virginia	0
Mississippi	0	Wisconsin	0
Missouri	0	Wyoming	0

Trend Check and Treatment Effects

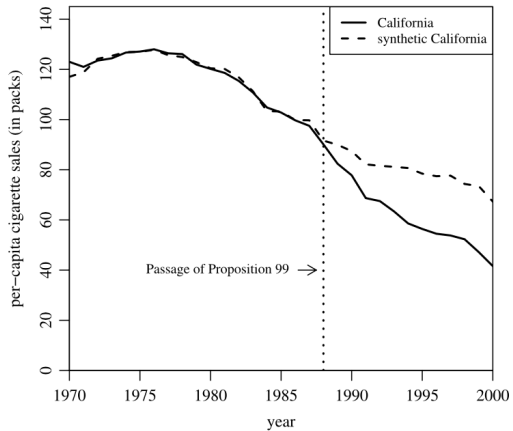
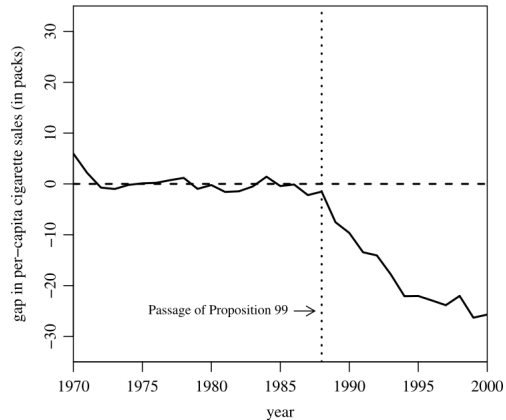


Figure 2. Trends in per-capita cigarette sales: California vs. synthetic California.



Theory

We're going to break this down into two parts:

1. How do we estimate w_j ?
2. Conditional on $\hat{w}_j$'s, how do we estimate the treatment effect?

A Starting Point

Start with $T_{it} = 1$ IFF $i = 1$ and $t > T_0$:

$$Y_{it} = Y_{it}(0) + \alpha_{it}T_{it}$$

Estimate $\alpha_1 = (\alpha_{1,T_0+1}, \alpha_{1,T_0+2}, \dots, \alpha_{1,T})$ (period by period treatment effect).

For $t > T_0$:

$$\alpha_{1t} = Y_{1t}(1) - Y_{1t}(0) = \underbrace{Y_{1t}}_{\text{observed}} - \underbrace{Y_{1t}(0)}_{\text{counterfactual}}$$

$$Y_{1t}(0) \approx \sum_{j=2}^{J+1} w_j Y_{jt}(0)$$

$$\hat{\alpha}_{1t} = Y_{1t}(1) - \sum_{j=2}^{J+1} w_j Y_{jt}(0) \quad \text{for } t = T_0 + 1, \dots, T$$

A Starting Point

Consider the model

$$Y_{it}(0) = \gamma_t + \theta_t \mathbf{Z}_i + \lambda_t \mathbf{F}_i + u_{it}$$

- δ_t is the time fixed effect
- Z_i are the usual covariates (fixed in i over time) with potentially time varying coefficients.
- F_i are i specific **unobserved factors**.
- λ_t are called **factor loadings**.

Some Algebra

So we define the synthetic observation as:

$$\sum_{j=2}^{J+1} w_j \underbrace{Y_{jt}(0)}_{\text{observed}} = \delta_t + \theta_t \left(\sum_{j=2}^{J+1} w_j \mathbf{Z}_j \right) + \lambda_t \left(\sum_{j=2}^{J+1} w_j \mathbf{F}_j \right) + \sum_{j=2}^{J+1} w_j u_{jt}$$

And the difference in untreated observations:

$$\begin{aligned} & Y_{1t}(0) - \sum_{j=2}^{J+1} w_j Y_{jt}(0) \\ &= \underbrace{\theta_t \left(\mathbf{Z}_1 - \sum_{j=2}^{J+1} w_j \mathbf{Z}_j \right)}_{\text{match this directly}} + \underbrace{\lambda_t \left(\mathbf{F}_1 - \sum_{j=2}^{J+1} w_j \mathbf{F}_j \right)}_{\text{match indirectly: } (Y_{i,1}, \dots, Y_{i,T_0})} + \sum_{j=2}^{J+1} w_j \underbrace{(u_{1t} - u_{jt})}_{\text{Usual } E[u_{it} | \mathbf{Z}_{it}, \mathbf{F}_{it}] = 0} \end{aligned}$$

Some Algebra

The goal is to match:

$$\mathbf{Z}_1 = \sum_{j=2}^{J+1} w_j \mathbf{Z}_j, \quad \mathbf{F}_1 \approx \sum_{j=2}^{J+1} w_j \mathbf{F}_j$$

but $\mathbf{F}$ is not observed.

- The main result in the Abadie et. al (2010) paper proves that the above equation holds approximately if

$$(Y_{1,1}, \dots, Y_{1,T_0}) = \sum_j w_j \cdot (Y_{j,1}, \dots, Y_{j,T_0})$$

The Problem

Theory depends on matching:

$$\mathbf{Z}_1 = \sum_{j=2}^{J+1} w_j \mathbf{Z}_j, \quad (Y_{1,1}, \dots, Y_{1,T_0}) = \sum_j w_j \cdot (Y_{j,1}, \dots, Y_{j,T_0})$$

- There may be no $w_2, \dots, w_J$ that satisfy the equations
- Instead choose $\mathbf{w}$ in a **minimum distance sense**.

The Problem

$$\mathbf{X}_i = (\mathbf{Z}_i, Y_{i,1}, \dots, Y_{i,T_0})$$
$$\arg \min_{w_2, \dots, w_J} (\mathbf{X}_i - \sum_j w_j \mathbf{X}_j)' \Omega (\mathbf{X}_i - \sum_j w_j \mathbf{X}_j) \quad \text{s.t.} \quad \sum_j w_j = 1$$

- Can choose Ω to minimize variance of the resulting estimator.
- Can get negative weights: what do those mean?
- We can rule them out $w_j \geq 0$ constraints but the problem becomes more difficult to solve.

Placebos and inference

Placebos I

- Placebo of regions
 - Iteratively apply the SCM to every other control region, shifting the treated region to the donor pool.
- Steps
 - Put California in the donor pool.
 - Pick a state from the donor pool at pretend that receives the treatment after T_0
 - Choose w_j following the synthetic control procedure.
 - Compute the treatment effects in the same way.
 - Repeat for all states in donor pool.

Ideally, the estimated effect in the treated region should be larger than the estimated effect for any other province not exposed.

Placebo of regions

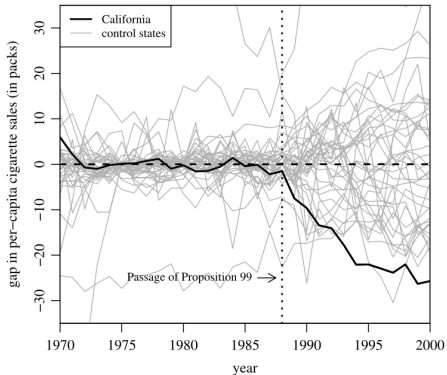


Figure 4. Per-capita cigarette sales gaps in California and placebo gaps in all 38 control states.

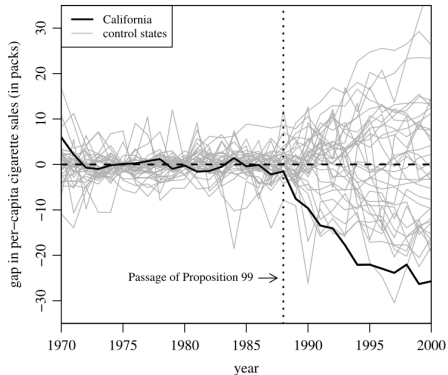


Figure 5. Per-capita cigarette sales gaps in California and placebo gaps in 34 control states (discards states with pre-Proposition 99 MSPE twenty times higher than California's).

Placebo of regions

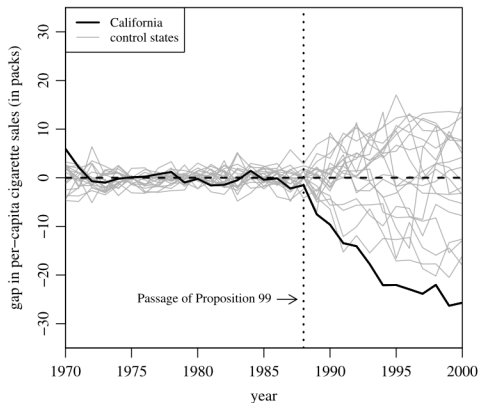


Figure 7. Per-capita cigarette sales gaps in California and placebo gaps in 19 control states (discards states with pre-Proposition 99 MSPE two times higher than California's).

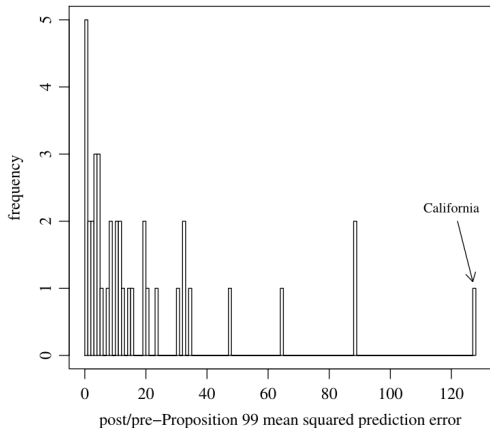


Figure 8. Ratio of post-Proposition 99 MSPE and pre-Proposition 99 MSPE: California and 38 control states.

The previous placebo can be used for inference.

- Basic idea: confidence in the validity of the estimated gap would disappear if the methodology also estimates large effects when applied iteratively to (permutations to generate placebos).
- The p-values can be constructed by computing the proportion of placebo gaps equal to or greater than the estimated gap for the treated unit:

$$p - value = Pr(\hat{\alpha}_{placebo} > \hat{\alpha}_1) = \frac{1}{J+1} \sum_{j=1}^{J+1} I(\hat{\alpha}_{jt}^{placebo} > \hat{\alpha}_{1t})$$

- In-time placebo
 - Randomly reassign the time when the intervention took place. Ideally, no impacts will be found in the pre-treatment period.
- Leave-out test
 - Iteratively apply the SCM omitting in each iteration, one of each region that received a positive weight and to rest of the provinces in the donor pool.
- Check the sensitivity of the results to the V weights.

When to use it...

- If the country or region is **not** conducting multiple reforms/projects at once
 - To be clear what impact is being isolated/identified
- When you analyze a region/state within a country
 - External shocks are more likely to be similar through regions within a country
- More credible if there are more pre-intervention periods
 - Maybe 10-20 instead of 5: you want to see that the movements are similar across business cycles
- More credible and accurate with more post-treatment periods
 - Perhaps 3 instead of 1: it is expected to see the gap increase or maintained years after the implementation of the reform.

Synthetic Difference in Differences

$$N = N_0 + N_1$$

$$T = T_0 + T_1$$

- N_0 is the amount of control units.
- N_1 is the amount of treated units.
- $t \leq T_0$ is the pre-treatment period.
- $t > T_0$ is the post-treatment period.

Synthetic Difference-in-Differences generalizes DID and SC

2 × 2 diff-in-diff

	pre-treatment	post-treatment
control	$\sum_{i \leq N_0} \sum_{t \leq T_0} \hat{\omega}_i \hat{\lambda}_t Y_{it}$	$\sum_{i \leq N_0} \sum_{t > T_0} \hat{\omega}_i T_1^{-1} Y_{it}$
treated	$\sum_{i > N_0} \sum_{t \leq T_0} N_1^{-1} \hat{\lambda}_t Y_{it}$	$\sum_{i > N_0} \sum_{t > T_0} N_1^{-1} T_1^{-1} Y_{it}$

Caso 1: SC

- SC takes only one difference *post-treatment*.
- Uses zero time weights $\lambda_t = 0$. Has unit weights $\hat{\omega}_i$.
- Has $N_1 = 1$, one treated region. **Assumption:** $T_1 = 1$, so $T = T_0 + 1$.

	pre-treatment	post-treatment (effect in $T_0 + 1$)
control	0	$\sum_{i \leq N_0} \hat{\omega}_i Y_{i,T_0+1}$
treated	0	Y_{1,T_0+1}

$$SC = Y_{1,T_0+1} - \sum_{i \leq N_0} \hat{\omega}_i Y_{i,T_0+1}$$

Caso 2: DiD

- DID uses unit weights and time weights but are **equal weights!**
- $\omega_i = 1/N_0$, $\lambda_t = 1/T_0$.
- **Assumption:** $T_1 = 1$, so $T = T_0 + 1$.

	pre-treatment	post-treatment (effect in $T_0 + 1$)
control	$\sum_{i \leq N_0} \sum_{t \leq T_0} \frac{Y_{it}}{N_0 T_0}$	$\sum_{i \leq N_0} \frac{Y_{it}}{N_0}$
treated	$\sum_{i > N_0} \sum_{t \leq T_0} \frac{Y_{it}}{N_1 T_0}$	$\sum_{i > N_0} \frac{Y_{it}}{N_1}$

$$DD_{T_0+1} = \left(\bar{Y}_{T_0+1}^{treated} - \bar{Y}_{T_0+1}^{control} \right) - \left(\bar{Y}_{pre}^{treated} - \bar{Y}_{post}^{pre} \right)$$

Caso 3: SDID

- SDID uses unit weights and time weights but allows **unequal weights**!
- $\omega_i \neq 1/N_0$, $\lambda_t \neq 1/T_0$.
- Time weights: they define a **pre-treatment period**.
- Unit weights: they define a **synthetic control regions**.

	pre-treatment	post-treatment
control	$\bar{Y}_{pre}^{control}$ (units and time weighted)	$\bar{Y}_{T_0+1}^{control}$ (units weighted)
treated	$\bar{Y}_{pre}^{treated}$ (time weight)	$\bar{Y}_{T_0+1}^{treated}$

$$SDID = \left(\bar{Y}_{T_0+1}^{treated} - \underbrace{\bar{Y}_{T_0+1}^{\omega_i > 0}}_{\text{Control defined by } \omega_i} \right) - \underbrace{\left(\bar{Y}_{\lambda_t > 0}^{treated} - \bar{Y}_{\lambda_t > 0}^{\omega_i > 0} \right)}_{\text{Pre period defined by } \lambda_t}$$

Estimating Unit Weights

1. We estimate unit weights defining the **synthetic control unit** on the pre-treatment data.

$$\hat{\omega} = \arg \min_{\omega_0, \omega} \left\| \omega_0 + \omega^T Y_{co,pre} - Y_{tr,pre} \right\|^2 + \zeta^2 T_{pre} \|\omega\|^2$$

subject to

$$\omega_i \geq 0, \quad \sum_{i=1}^{N_{co}} \omega_i = 1.$$

It allows a constant term ω_0 ! Difference with synthetic control $\Rightarrow$ synthetic and treated parallel trends.

Estimating Time Weights

- 2 We estimate the time weights defining the **synthetic pre-treatment period** on the control data.

$$\hat{\lambda} = \arg \min_{\lambda_0, \lambda} \|\lambda_0 + Y_{co,pre}\lambda - Y_{co,post}\|^2$$

subject to

$$\lambda_t \geq 0, \quad \sum_{t=1}^{T_{pre}} \lambda_t = 1.$$

We impose analogous constraints.

Last: With λ_t and ω_i go back to table, and just DiD.

Consider W_{it} treatment dummy, and no controls in the regression:

- **SC** is **weighted** linear regression, no unit fe's (slightly unconventional representation, but identical to standard one):

$$\hat{\tau}^{SC} = \operatorname{argmin}_{\tau, \gamma} \sum_{i,t} (Y_{it} - \gamma_t - \tau W_{it})^2 \times \omega_i^{SC}$$

- **DID** is **unweighted** regression with unit and time fe's:

$$\hat{\tau}^{DID} = \operatorname{argmin}_{\tau, \gamma, \alpha} \sum_{i,t} (Y_{it} - \gamma_t - \alpha_i - \tau W_{it})^2$$

- **Synthetic Difference In Differences (SDID)** is **weighted** regression with unit and time fe's:

$$\hat{\tau}^{SDID} = \operatorname{argmin}_{\tau, \gamma, \alpha} \sum_{i,t} (Y_{it} - \gamma_t - \alpha_i - \tau W_{it})^2 \times \omega_i \times \lambda_t$$

Application to California smoking data

- Abadie-Diamond-Hainmueller 2010 JASA paper application.
- California starts anti-smoking program in 1989.
- Outcome is per capita cigarette sales (packs/year).

Our approach

- Estimate unit and time weights.
- Run weighted DID regression.

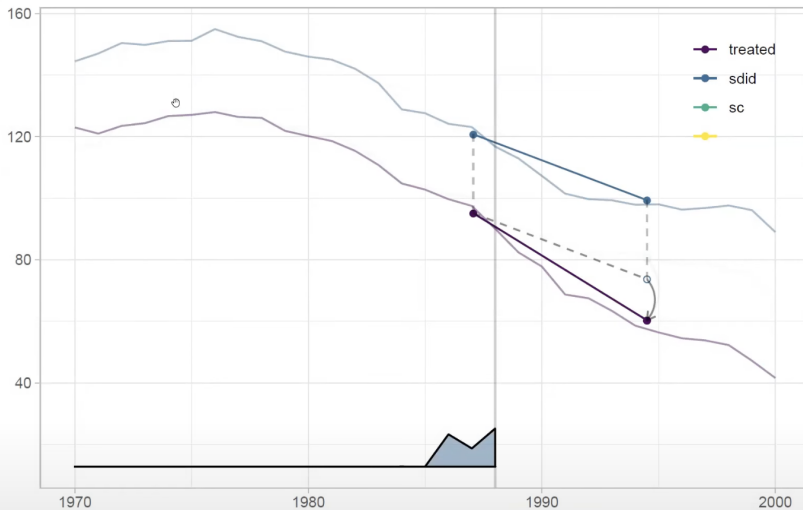


Figure 1: Sdid

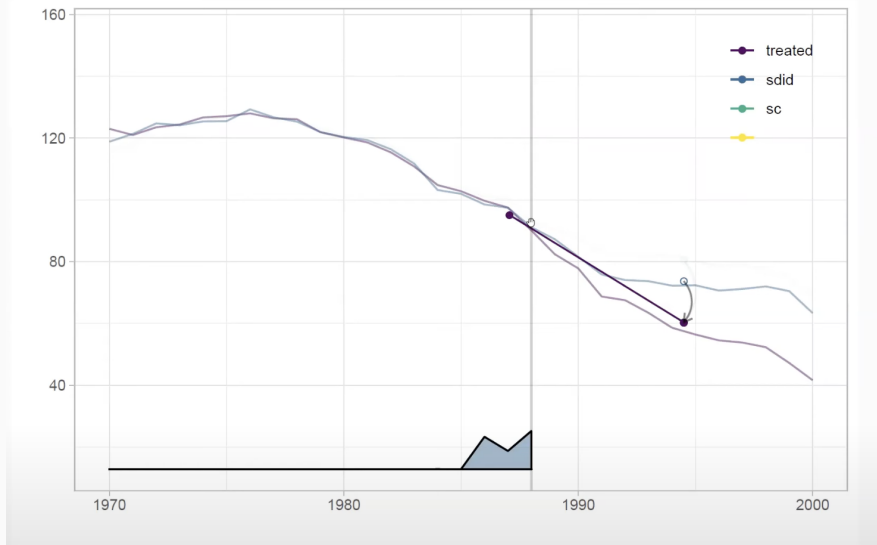


Figure 2: Sdid (no constant)

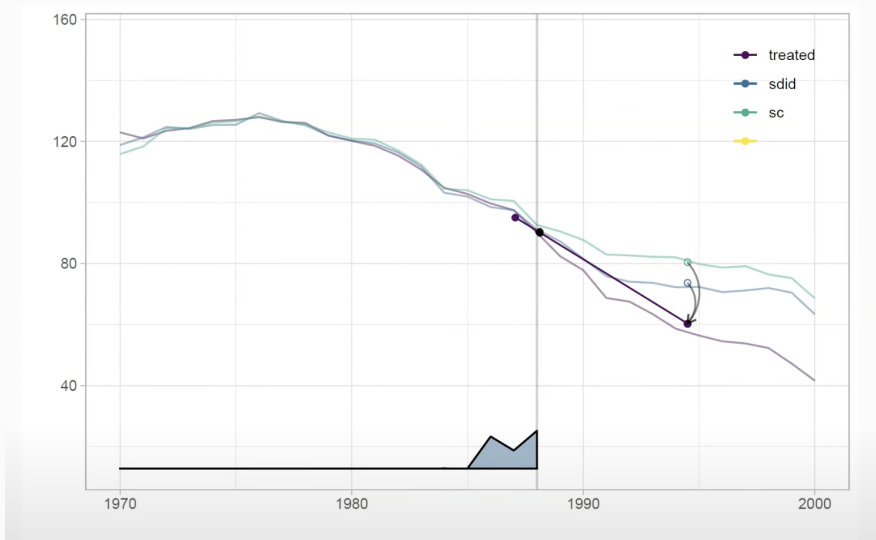


Figure 3: Synthetic control

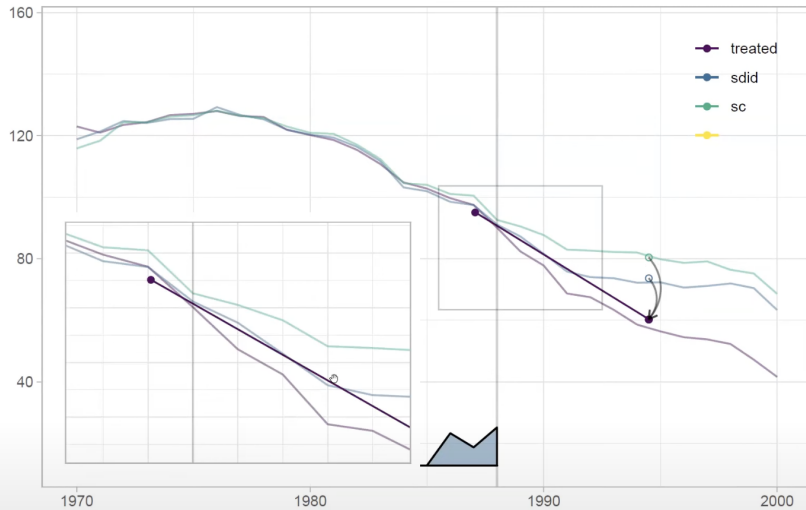


Figure 4: Synthetic control (zoom)

Unit Weights ω

- 24 of 37 states have zero weights for both SDID and SC.

state	weight_sdid	weight_sc
Colorado	0.07	0.07
Connecticut	0.13	0.07
Delaware	0.07	—
Idaho	—	0.05
Illinois	0.09	—
Kansas	0.03	—
Minnesota	0.04	—
Montana	0.04	0.14
Nebraska	0.06	0.01
Nevada	0.18	0.17
New Hampshire	0.10	0.03
New Mexico	0.04	0.12
Utah	0.04	0.28
Wisconsin	0.03	—

Time weights λ

- All but the last three pre-treatment periods have zero weight.
- SDID Regression of $\bar{Y}_{i,a}$ on $Y_{i,1}, \dots, Y_{i,T_0}$ has $R^2 = 0.871$.

year	weight
1970	—
$\vdots$	—
1985	—
1986	0.36
1987	0.20
1988	0.43

- SDID focuses on 13 states, and last 3 time periods.

	DID	SC	SDID
Estimate	-27.4	-20.1	-13.4
(SE)	(16.4)	(7.7)	(7.6)

- Even though SC fit in the pre-treatment period is good, there are some deviations in the last three periods that make the SDID estimate deviate from the SC estimate.
- DID implicitly has poor fit in the pre-treatment period.